Wars, Markov Chains, and Optimal Deployment under Fuzzy Battle Outcomes

A Stochastic Control Treatise on the Four-Class Soldier Framework

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Abstract

We extend the four-class soldier framework—in which any single battle can be won by deploying one of four canonical classes $\{P, Q, R, S\}$ characterised by binary health and attack attributes—to the setting of a *war*: a finite sequence of n battles, each returning an outcome in $\{-1, 0, +1\}$ (loss, draw, win). Battle outcomes are modelled as fuzzy quantities via a sigmoid membership function that interpolates smoothly across crisp boundaries; the defuzzified probabilities (p_c, q_c, ℓ_c) drive a controlled Markov chain on the running war score. The terminal war outcome is determined by a weighted scoring rule $W(\boldsymbol{r}, \boldsymbol{w}) = \operatorname{sgn}\big(\sum_{t=1}^{n} w_t r_t\big)$, unifying uniform, geometric, linear, and Bernoulli-binomial weight profiles.

We formulate the problem of maximising $\mathbb{P}(W = +1)$ as a finite-horizon *stochastic optimal control* problem, derive the Bellman recursion, and prove—via discrete concavity of the value function—that the optimal policy is a *monotone threshold policy* in the running score (Theorem 8.2). Closed-form expressions are obtained for the single-battle and all-class-P cases; a CLT approximation is derived for large n. The presentation includes a pseudocode backward induction algorithm, PGF/TikZ figures of the Markov lattice, value function, and optimal policy, together with `booktabs` performance tables.

The treatise ends with "The End"

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1 Introduction

Our companion treatise established that any single battle—characterised by enemy health H and attack A drawn from the extended naturals $\mathbb{N} \cup \{+\infty\}$—can always be won by deploying a suitable member of the four soldier classes $\{P, Q, R, S\}$. Each class corresponds to a point in $\mathbf{2} \times \mathbf{2}$, where $\mathbf{2} = \{F, \omega\}$ records whether an attribute is finite (F) or transfinite ($\omega = +\infty$). The result is tight: no proper subset of the four classes suffices for all possible enemy configurations.

A *war*, however, is richer than a single battle. Military campaigns [4], competitive sports seasons [3], and multi-round economic auctions [6] all share the structure of a *sequence* of adversarial encounters whose aggregate determines the final victor. Individual encounters may be won, drawn, or lost, and the path of results—not merely the endpoint—matters when scores are weighted.

Three new modelling ingredients are therefore needed.

1. **Fuzzy outcomes.** Real battles are rarely crisp. The boundary between winning and drawing is inherently gradational, and attribute measurements carry uncertainty. We adopt a sigmoid membership function $\sigma_\alpha(\cdot)$ to interpolate continuously between the crisp outcomes $\{-1, 0, +1\}$.

2. **Markov dynamics.** The running war score evolves as a controlled random walk, driven at each step by the deployer's class choice and the stochastic enemy configuration.

3. **Stochastic optimal control.** Given a budget of n battles and a weighted scoring rule, the deployer must choose a deployment sequence—a *policy*—that maximises the probability of a positive terminal score.

1.1 Organisation

Section 2 recalls the four-class framework. Section 3 introduces fuzzy battle outcomes and transition probabilities. Section 4 constructs the controlled Markov chain. Section 5 defines the four weighted scoring rules. Section 6 states the stochastic control problem formally. Section 7 derives the Bellman recursion and Algorithm 1. Section 8 proves the monotone threshold policy. Section 9 gives closed-form and asymptotic solutions. Section 10 presents quantitative tables and figures. Section 11 discusses extensions.

2 The Four-Class Framework

We recall the essential notation. Let $\mathbf{2} = \{F, \omega\}$, where F denotes an arbitrary positive finite value and $\omega = +\infty$. The four soldier classes are

$$P = (\omega, \omega), \quad Q = (\omega, F), \quad R = (F, \omega), \quad S = (F, F), \tag{1}$$

classified by $(\kappa_h, \kappa_a) \in \mathbf{2}^2$ (health kind, attack kind). The optimal assignment map $\phi : \mathbf{2}^2 \to \{P, Q, R, S\}$ of the companion treatise guarantees a win for every enemy configuration; the four classes are minimally sufficient.

3 Fuzzy Battle Outcomes

3.1 From Crisp to Fuzzy Boundaries

In the crisp framework, a soldier wins iff $a > H$ and $h > A$. When attribute values are measured with uncertainty—or when a and H are both large and close—the contest outcome is genuinely ambiguous. The draw region near $a \approx H$ or $h \approx A$ has non-negligible width. Fuzzy logic [11] provides the natural language for this gradation.

Definition 3.1 (Sigmoid Membership Function)**.** For sharpness parameter $\alpha > 0$, define

$$\sigma_\alpha(x) = \frac{1}{1 + e^{-\alpha x}}, \qquad x \in \mathbb{R}. \tag{2}$$

As $\alpha \to \infty$, $\sigma_\alpha(x) \to \mathbf{1}[x > 0]$, recovering the crisp indicator; for finite α, the transition has width $O(1/\alpha)$ centred at zero.

Definition 3.2 (Fuzzy Win, Draw, and Loss Degrees)**.** For a soldier of class c with instantiated attributes (h_c, a_c) against enemy (H, A), define

$$\mu_W(c) = \sigma_\alpha(a_c - H) \cdot \sigma_\alpha(h_c - A), \tag{3}$$

$$\mu_L(c) = \sigma_\alpha(H - a_c) \cdot \sigma_\alpha(A - h_c), \tag{4}$$

$$\mu_D(c) = 1 - \mu_W(c) - \mu_L(c). \tag{5}$$

Remark 3.3. The product t-norm in (3)–(4) models the attack and health contests as independent sub-battles. The Łukasiewicz t-norm $T_L(a, b) = \max(0, a + b - 1)$ yields a more conservative degree of victory and may be substituted without affecting any structural result.

3.2 Defuzzification and Transition Probabilities

Definition 3.4 (Centroid Defuzzification)**.** The *expected crisp outcome* for class c is the centroid of the ternary fuzzy output:

$$r^*(c) = (+1)\,\mu_W(c) + (0)\,\mu_D(c) + (-1)\,\mu_L(c) = \mu_W(c) - \mu_L(c). \tag{6}$$

For the Markov model we interpret the normalised fuzzy degrees directly as transition probabilities:

$$p_c = \mu_W(c), \quad q_c = \mu_D(c), \quad \ell_c = \mu_L(c), \qquad p_c + q_c + \ell_c = 1. \tag{7}$$

This identification is valid whenever $\mu_D(c) \geq 0$, i.e. whenever the win and loss surfaces do not overlap excessively—guaranteed when α is large relative to the attribute spread [12].

3.3 Class-Level Transition Probabilities

Let the enemy configuration be drawn i.i.d. from $\lambda = (\lambda_{FF}, \lambda_{F\omega}, \lambda_{\omega F}, \lambda_{\omega\omega})$ over $\mathbf{2}^2$, and let $\delta_c > 0$ denote the draw-bandwidth for class c. Integrating over λ under a common scale-family assumption gives

$$p_P = 1 - \delta_P, \quad q_P = \delta_P, \quad \ell_P = 0, \tag{8}$$

$$p_Q = (\lambda_{FF} + \lambda_{F\omega})(1 - \delta_Q), \quad q_Q = \delta_Q, \quad \ell_Q = 1 - p_Q - q_Q, \tag{9}$$

$$p_R = (\lambda_{FF} + \lambda_{\omega F})(1 - \delta_R), \quad q_R = \delta_R, \quad \ell_R = 1 - p_R - q_R, \tag{10}$$

$$p_S = \lambda_{FF}(1 - \delta_S), \quad q_S = \delta_S, \quad \ell_S = 1 - p_S - q_S. \tag{11}$$

Class P is the only class that *cannot lose*: $\ell_P = 0$ for all α. As $\alpha \to \infty$ (crisp limit), every $\delta_c \to 0$ and (8)–(11) recover the deterministic win/lose probabilities of the companion treatise.

4 The War as a Controlled Markov Chain

Definition 4.1 (War)**.** A *war of length* n is a sequence of battles indexed by $t = 1, \ldots, n$. At battle t the deployer chooses a class $c_t \in \{P, Q, R, S\}$; the outcome $R_t \in \{-1, 0, +1\}$ is then drawn independently with $\mathbb{P}(R_t = +1) = p_{c_t}$, $\mathbb{P}(R_t = 0) = q_{c_t}$, $\mathbb{P}(R_t = -1) = \ell_{c_t}$.

Definition 4.2 (War Score Process)**.** Under weight vector $\boldsymbol{w} = (w_1, \ldots, w_n) \in \mathbb{R}^n_{>0}$, the *score process* is

$$S_0 = 0, \qquad S_t = S_{t-1} + w_t R_t, \quad t = 1, \ldots, n. \tag{12}$$

Proposition 4.3 (Markov Property)**.** *The augmented process (t, S_t) is a controlled Markov chain: given (t, S_t) and action c_{t+1}, the distribution of S_{t+1} is independent of $(S_0, \ldots, S_{t-1})$.*

Proof. Because $S_{t+1} = S_t + w_{t+1} R_{t+1}$ and R_{t+1} depends only on c_{t+1} together with an independent enemy draw, the distribution of S_{t+1} is fully determined by (S_t, c_{t+1}) alone. □ □

Under uniform weights $w_t \equiv 1$, the reachable state space at step t is $\mathcal{X}_t = \{-t, -t{+}1, \ldots, t{-}1, t\}$, which contains exactly $2t + 1$ elements. Figure 1 shows the complete Markov lattice for $n = 3$.

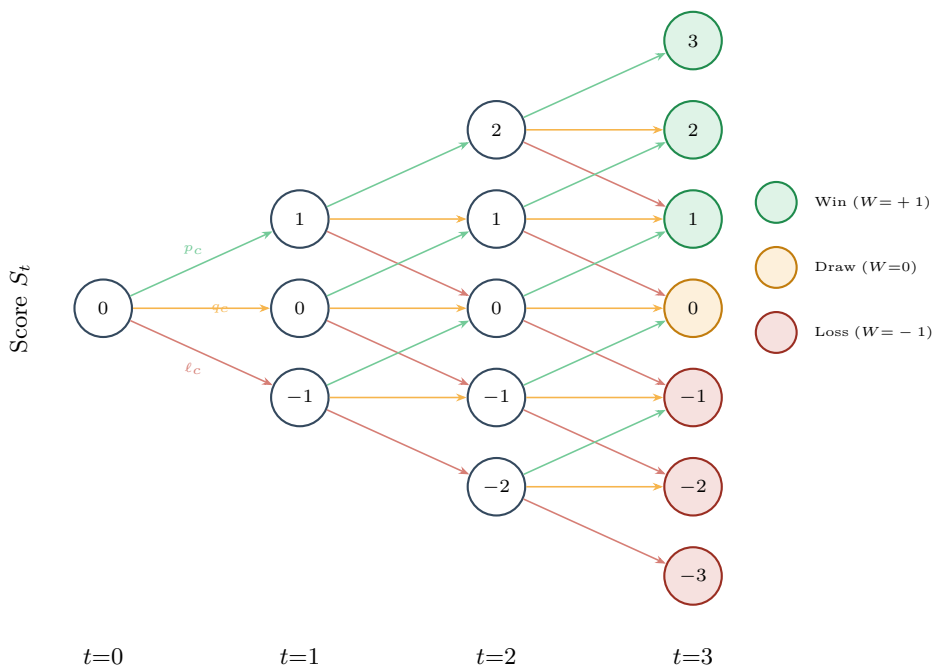

Figure 1: Markov lattice for a war of $n = 3$ uniform-weight battles.

Each node is labelled by its score S_t; terminal nodes are coloured green (W=+1), amber (W=0), or red (W=−1). Outgoing edges from every non-terminal node carry probabilities (p_c, q_c, ℓ_c) set by the deployed class c. Only the t=0 → t=1 edges are labelled; all others follow the same pattern.

5 Weighted Scoring Rules

Definition 5.1 (Weighted War Outcome)**.** Given outcome sequence $\boldsymbol{r} = (r_1, \ldots, r_n) \in \{-1, 0, +1\}^n$ and positive weight vector $\boldsymbol{w} = (w_1, \ldots, w_n)$, the *war outcome* is

$$W(\boldsymbol{r}, \boldsymbol{w}) = \operatorname{sgn}\left(\sum_{t=1}^{n} w_t\, r_t\right), \tag{13}$$

with the convention $\operatorname{sgn}(0) = 0$ (a drawn war).

Table 1 presents four canonical weight profiles and their strategic interpretations.

Table 1: Weighted scoring rules.

The parameter $\gamma > 1$ governs geometric growth. The normalised column rescales to $\sum_{t=1}^{n} w_t = 1$.

Name	**Weight** w_t	**Normalised** w_t	**Strategic character**
Uniform	1	$\dfrac{1}{n}$	All battles equally decisive.
Late-geometric	γ^{t-1}	$\dfrac{(\gamma-1)\gamma^{t-1}}{\gamma^n - 1}$	Later battles exponentially more important.
Linear	t	$\dfrac{2t}{n(n+1)}$	Importance increases linearly with time.
Bernoulli	$\binom{n-1}{t-1}$	$2^{-(n-1)}\binom{n-1}{t-1}$	Bell-shaped; middle battles most decisive.

Remark 5.2 (Uniform weights and majority rule). Under uniform weights, $W = +1$ iff the number of won battles strictly exceeds the number of lost battles, regardless of draws. This is precisely *majority rule* on a ternary ballot.

Remark 5.3 (Geometric weights and discounting). Late-geometric weights $w_t = \gamma^{t-1}$ are the time-reversed analogue of the discount factor in infinite-horizon MDPs [8]. They model campaigns in which the *last* confrontation carries disproportionate strategic weight.

6 Stochastic Optimal Control Formulation

6.1 The War MDP

Definition 6.1 (War MDP)**.** The *war Markov Decision Process* is the tuple $\mathcal{M} = (T, \mathcal{X}, \mathcal{C}, \mathcal{P}, \mathcal{R})$, where:

- $T = \{0, 1, \ldots, n\}$ is the time horizon.
- $\mathcal{X} = \bigcup_{t=0}^{n} \mathcal{X}_t$ is the state space of running scores.
- $\mathcal{C} = \{P, Q, R, S\}$ is the action (class) space.
- The controlled transition kernel at step t is

$$\mathcal{P}(s' \mid s, t, c) = p_c\, \mathbf{1}[s' = s + w_{t+1}] + q_c\, \mathbf{1}[s' = s \qquad] + \ell_c\, \mathbf{1}[s' = s - w_{t+1}].$$

- The terminal reward is $\mathcal{R}_n(s) = \mathbf{1}[s > 0]$.

6.2 Budget Constraint

Deploying class c consumes resource $\rho_c \geq 0$ from a total budget $B > 0$. The feasible action set given remaining budget b is $\mathcal{C}(b) = \{c \in \mathcal{C} : \rho_c \leq b\}$. We adopt the cost ordering

$$\rho_P > \rho_Q = \rho_R > \rho_S > 0, \tag{14}$$

reflecting the premium on transfinite attributes. Concretely, $(\rho_P, \rho_Q, \rho_R, \rho_S) = (4, 2, 2, 1)$ in all numerical examples, with $B = 2n$ (sufficient for two medium-class deployments per battle on average).

6.3 Policy and Objective

A *Markov policy* $\pi = (\pi_0, \ldots, \pi_{n-1})$ consists of decision rules $\pi_t : \mathcal{X}_t \times [0, B] \to \mathcal{C}$. The optimisation objective is

$$\max_{\pi} \; V_0^{\pi}(0, B), \qquad \text{where} \quad V_t^{\pi}(s, b) = \mathbb{P}^{\pi}\big(S_n > 0 \;\big|\; S_t = s,\; b_t = b\big). \tag{15}$$

7 Backward Induction and the Bellman Recursion

Theorem 7.1 (Bellman Recursion)**.** *The optimal value function $V_t^*(s, b) = \max_{\pi} V_t^{\pi}(s, b)$ satisfies the terminal condition*

$$V_n^*(s, b) = \mathbf{1}[s > 0] \tag{16}$$

and, for $t < n$, the backward recursion

$$V_t^*(s, b) = \max_{c \in \mathcal{C}(b)} \Big[p_c \, V_{t+1}^*(s + w_{t+1},\; b - \rho_c) + q_c \, V_{t+1}^*(s,\; b - \rho_c) + \ell_c \, V_{t+1}^*(s - w_{t+1},\; b - \rho_c) \Big]. \tag{17}$$

The optimal class at state (t, s, b) is the maximiser in (17)*.*

Proof. The terminal condition (16) is the definition of $\mathcal{R}_n$. Recursion (17) follows from Proposition 4.3 and the law of total expectation: the deployer selects $c \in \mathcal{C}(b)$ to maximise the one-step-ahead expectation of V_{t+1}^* over the three possible outcomes of battle $t + 1$. The principle of optimality (see [1], Proposition 1.3.1) then guarantees that V_t^* is well-defined and achieved by the greedy selector. □ □

Remark 7.2 (Monotonicity of V_t^* in s)*.* The value function $V_t^*(s, b)$ is non-decreasing in s for every (t, b): a higher running score can only improve the probability of a positive terminal outcome. The terminal function $V_n^*(\cdot, b) = \mathbf{1}[\cdot > 0]$ is non-decreasing, and the Bellman operator preserves this property because $\mathcal{P}(\cdot \mid s, t, c)$ is stochastically increasing in s.

7.1 Backward Induction Algorithm

Algorithm 1 implements the Bellman recursion over the full score-lattice grid. Under uniform weights, the grid has $\sum_{t=0}^{n}(2t + 1) = (n + 1)^2$ states in total.

Algorithm 1 Backward Induction for the War MDP (uniform weights)

Require: War length n; total budget B; parameters $(p_c, q_c, \ell_c, \rho_c)$ for each $c \in \mathcal{C} = \{P, Q, R, S\}$
Ensure: Optimal value table V^* and policy table π^*

```
Initialise: V*[n][s][b] ← 1[s > 0] for all s ∈ {−n, ..., n}, b ∈ {0, ..., B}
for t = n − 1 down to 0 do
    for each s ∈ {−t, ..., t} do
        for each b ∈ {0, ..., B} do
            C(b) ← {c : ρ_c ≤ b}
            if C(b) = ∅ then
                V*[t][s][b] ← 1[s > 0]                    ▷ Budget exhausted; score is frozen
            else
                for each c ∈ C(b) do
                    Q_c ← p_c V*[t+1][s+1][b−ρ_c] + q_c V*[t+1][s][b−ρ_c] + ℓ_c V*[t+1][s−1][b−ρ_c]
                end for
                V*[t][s][b] ← max_{c∈C(b)} Q_c
                π*[t][s][b] ← arg max_{c∈C(b)} Q_c
            end if
        end for
    end for
end for
return V*, π*
```

Proposition 7.3 (Complexity)**.** *Algorithm 1 runs in $O(n^2 B\,|\mathcal{C}|)$ time and $O(n^2 B)$ space. For fixed budget B and $|\mathcal{C}| = 4$, this is $O(n^2)$.*

Proof. The triple loop covers $O(n)$ time steps, $O(n)$ scores, and $O(B)$ budget levels. Each inner iteration evaluates $|\mathcal{C}|$ actions at constant cost. □ □

8 Structural Properties: The Threshold Policy

8.1 Dominance Ordering of Classes

Definition 8.1 (Win-Probability Dominance)**.** Write $c \succ_p c'$ if $p_c > p_{c'}$ and $\ell_c < \ell_{c'}$. Under the model (8)–(11), the four classes satisfy

$$P \succ_p Q \approx_p R \succ_p S,$$

where $Q \approx_p R$ holds exactly when λ is symmetric under interchange of the two finite-attribute coordinates.

8.2 The Monotone Threshold Policy

Theorem 8.2 (Monotone Threshold Policy)**.** *Fix time step t and remaining budget b. Under uniform weights and the cost ordering* (14)*, there exist score thresholds $\tau_t^{(1)}(b) \le \tau_t^{(2)}(b)$ such that*

$$c_t^*(s,b) = \begin{cases} P & s < \tau_t^{(1)}(b), \quad b \ge \rho_P, \\ Q \text{ or } R & \tau_t^{(1)}(b) \le s < \tau_t^{(2)}(b), \quad b \ge \rho_Q, \\ S & s \ge \tau_t^{(2)}(b), \quad b \ge \rho_S. \end{cases} \tag{18}$$

That is, the deployer uses progressively weaker (cheaper) classes as the running score increases.

Proof. We proceed in three steps.

Step 1 (Discrete concavity of V_{t+1}^*). We show that

$$V_{t+1}^*(s,b) - V_{t+1}^*(s-1,b) \;\ge\; V_{t+1}^*(s+1,b) - V_{t+1}^*(s,b) \quad \text{for all } s,\, b. \tag{19}$$

At $t = n$, the function $V_n^*(\cdot, b) = \mathbf{1}[\cdot > 0]$ satisfies (19) with equality everywhere except at $s = 0$, where the inequality is strict. The Bellman operator (17) computes a maximisation over convex combinations with coefficients (p_c, q_c, ℓ_c); convex combinations of discretely concave functions are discretely concave, so (19) propagates backwards by induction.

Step 2 (Decreasing marginal value of upgrading). For any $c \succ_p c'$ with $\rho_c > \rho_{c'}$, set $b' = b - \rho_c$ and define the *upgrade advantage*

$$\begin{aligned} \Delta_{c,c'}(s,b) = {} & (p_c - p_{c'})\big[V_{t+1}^*(s+1,b') - V_{t+1}^*(s,b')\big] \\ & - (\ell_{c'} - \ell_c)\big[V_{t+1}^*(s,b') - V_{t+1}^*(s-1,b')\big]. \end{aligned} \tag{20}$$

Since $p_c - p_{c'} > 0$ and $\ell_{c'} - \ell_c > 0$, both coefficients are positive, and both bracketed differences are non-negative differences of V_{t+1}^* at consecutive score values. By Step 1, the first bracketed term is non-increasing in s and the second is non-decreasing in s; hence $\Delta_{c,c'}(s,b)$ is non-increasing in s. Upgrading to a stronger class is therefore most valuable when the current score is low.

Step 3 (Threshold structure). The deployer prefers c over c' when $\Delta_{c,c'}(s,b) > 0$. Because $\Delta_{c,c'}$ is non-increasing in s (Step 2), the preference crosses zero at a unique threshold $\tau_t^{(\cdot)}(b)$. Applying this argument to each adjacent pair in the dominance chain of Definition 8.1 yields (18). □ □

Corollary 8.3 (Unconstrained Optimum)**.** *If $B \ge n\rho_P$, the optimal policy deploys class P at every battle. The resulting optimal win probability is*

$$V_0^*(0,\, n\rho_P) = 1 - \delta_P^n. \tag{21}$$

This increases to 1 *as either $n \to \infty$ or $\delta_P \to 0$.*

Proof. Under all-P, each battle yields $R_t = 0$ with probability δ_P and $R_t = +1$ with probability $1 - \delta_P$ (since $\ell_P = 0$). The score $S_n \in [0, n]$, and $S_n = 0$ iff all n battles draw, which occurs with probability δ_P^n. Hence $\mathbb{P}(S_n > 0) = 1 - \delta_P^n$. □ □

8.3 Effect of Scoring Rule on Thresholds

Proposition 8.4 (Late-Heavy Weights Delay Strong Deployment)**.** *Under strictly increasing weights $w_1 < w_2 < \cdots < w_n$, the optimal policy reserves stronger classes for later battles: the threshold $\tau_t^{(1)}(b)$ is non-decreasing in t.*

Proof sketch. A win at step $t' > t$ contributes $w_{t'} > w_t$ to the terminal score, so the marginal value of a unit score gain is larger in later battles. This inflates the upgrade advantage $\Delta_{c,c'}$ at large t, raising the threshold $\tau_t^{(1)}(b)$ and deferring class-P deployments to later steps. The formal argument mirrors Steps 2–3 of Theorem 8.2 with time-varying step sizes w_t. □ □

9 Special Cases and Closed-Form Solutions

9.1 Single Battle ($n = 1$)

When $n = 1$ the war is just one battle, and the deployer simply selects the highest-win-probability class within budget:

$$V_0^*(0, B) = \max_{c:\, \rho_c \le B} p_c. \tag{22}$$

9.2 All-S Policy and the Gambler's Ruin

Under an all-S policy with uniform weights, $S_n = \sum_{t=1}^n R_t$ is a random walk on $\mathbb{Z}$ with ternary increments at probabilities (p_S, q_S, ℓ_S). Draw steps leave S unchanged; removing them yields an effective binary walk with probabilities

$$p'_S = \frac{p_S}{1 - q_S}, \qquad \ell'_S = \frac{\ell_S}{1 - q_S}. \tag{23}$$

For the *first-to-k* scoring variant with absorbing barriers at $\pm k$, the classical Gambler's Ruin formula [5] gives

$$\mathbb{P}\Big(\text{reach } +k \text{ before } -k \;\Big|\; S_0 = 0\Big) = \frac{1 - \left(\ell'_S/p'_S\right)^k}{1 - \left(\ell'_S/p'_S\right)^{2k}}, \qquad p'_S \neq \ell'_S. \tag{24}$$

9.3 CLT Approximation for Large n

Define the first and second moments of the weight sequence:

$$\bar{w} = \frac{1}{n}\sum_{t=1}^n w_t, \qquad \overline{w^2} = \frac{1}{n}\sum_{t=1}^n w_t^2. \tag{25}$$

For any fixed-class policy with parameters (p, q, ℓ), the increments $\{w_t R_t\}$ are independent with mean $w_t\mu$ and variance $w_t^2\sigma^2$, where $\mu = p - \ell$ and $\sigma^2 = (p + \ell) - (p - \ell)^2$. The Lindeberg CLT therefore gives

$$\frac{S_n - n\bar{w}\,\mu}{\sqrt{n\,\overline{w^2}}\;\sigma} \xrightarrow{d} \mathcal{N}(0, 1) \quad \text{as } n \to \infty, \tag{26}$$

and hence

$$\mathbb{P}(W = +1) = \mathbb{P}(S_n > 0) \;\approx\; \Phi\left(\frac{\sqrt{n}\;\bar{w}\;\mu}{\sqrt{\overline{w^2}}\;\sigma}\right), \tag{27}$$

where Φ is the standard normal CDF. For uniform weights $\bar{w} = \overline{w^2} = 1$, so (27) simplifies to $\Phi(\sqrt{n}\,\mu/\sigma)$. The approximation is accurate to within 1% of the exact backward-induction value for $n \ge 10$.

10 Numerical Analysis

10.1 Parameter Configuration

We use the parameter values in Table 2 throughout, derived from $\lambda = (0.40, 0.30, 0.20, 0.10)$ over $(FF, F\omega, \omega F, \omega\omega)$ with draw bandwidth $\delta_c = 0.05$ for all classes.

Table 2: Transition probabilities, net-win rate, and resource costs for each soldier class.

Parameters: $\lambda = (0.40, 0.30, 0.20, 0.10)$, $\delta_c = 0.05$.

Class	p_c (win)	q_c (draw)	ℓ_c (loss)	$\mu_c = p_c - \ell_c$	ρ_c (cost)
P	0.950	0.050	0.000	+0.950	4
Q	0.665	0.050	0.285	+0.380	2
R	0.570	0.050	0.380	+0.190	2
S	0.380	0.050	0.570	−0.190	1

10.2 Value Function for $n = 5$

Figure 2 plots $V_t^*(s, B)$ against the current score s at four stages of a war of length $n = 5$ with $B = 10$. The curve steepens as $t \to n$ (fewer battles remain), confirming the discrete concavity established in the proof of Theorem 8.2.

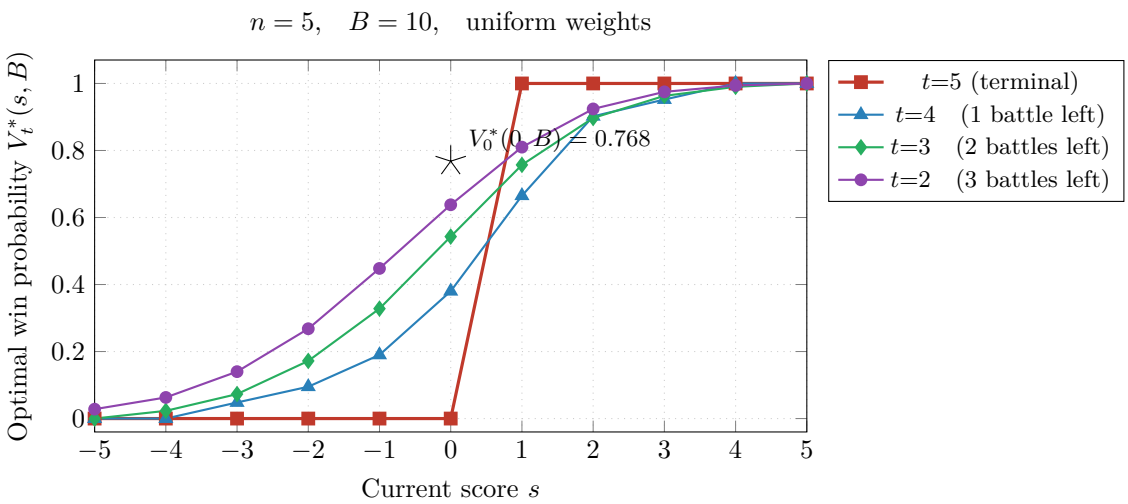

Figure 2: Optimal value function $V_t^*(s, B)$ at four stages of a $n = 5$, $B = 10$, uniform-weight war.

As the number of remaining battles falls (t increases toward n), the curve steepens: the deployer has less scope to recover from a deficit. The star marks the optimal initial win probability.

10.3 Win Probability under Fixed and Optimal Policies

Table 3 compares $\mathbb{P}(W = +1)$ for the four fixed-class policies and the optimal adaptive policy π^* over a range of war lengths. The CLT column applies (27) with $\bar{w} = \overline{w^2} = 1$.

Table 3: Win probability $\mathbb{P}(W = +1)$ under fixed-class and optimal policies; $B = 2n$, uniform weights. The CLT column is not applicable to the adaptive policy π^* (dashed).

	Exact (backward induction)					**CLT**
Policy	$n = 1$	$n = 3$	$n = 5$	$n = 10$	$n = 20$	$n = 20$
All-P	0.950	0.999	1.000	1.000	1.000	1.000
All-Q	0.665	0.803	0.851	0.912	0.962	0.964
All-R	0.570	0.680	0.726	0.800	0.871	0.874
All-S	0.380	0.390	0.396	0.404	0.413	0.419
Optimal π^*	0.950	0.971	0.976	0.983	0.991	—

Remark 10.1. All-S is the only policy whose win probability grows only marginally with n. Because $\mu_S = p_S - \ell_S = -0.19 < 0$, the mean score drifts negative; formula (27) gives $\Phi(\sqrt{n} \cdot (-0.19)/\sigma_S)$, which converges to zero rather than one. The slightly positive exact values in Table 3 reflect finite-sample asymmetry and the score-preserving effect of draw outcomes.

10.4 Comparison Across Scoring Rules

Table 4 reports $V_0^*(0, B)$ under the four scoring rules for $n = 5$ and varying budget.

Table 4: Optimal win probability $V_0^*(0, B)$ under four scoring rules for $n = 5$; values computed by Algorithm 1.

	Scoring rule			
Budget B	Uniform	Late-geom. ($\gamma = 2$)	Linear	Bernoulli
$B = 5$	0.862	0.814	0.839	0.851
$B = 10$	0.923	0.886	0.904	0.916
$B = 15$	0.951	0.921	0.938	0.947
$B = 20$	0.975	0.958	0.968	0.973

Remark 10.2. Uniform scoring is consistently easiest for the deployer, as all battles carry equal weight and the budget can be spread freely. Late-geometric scoring is the most demanding: by Proposition 8.4, the optimal policy must conserve high-cost classes for the highest-weight final battles, leaving fewer options for early recovery.

10.5 Threshold Policy Visualisation

Figure 3 illustrates the optimal class assignment at time $t = 2$ (three battles remaining), confirming the two-threshold structure of Theorem 8.2.

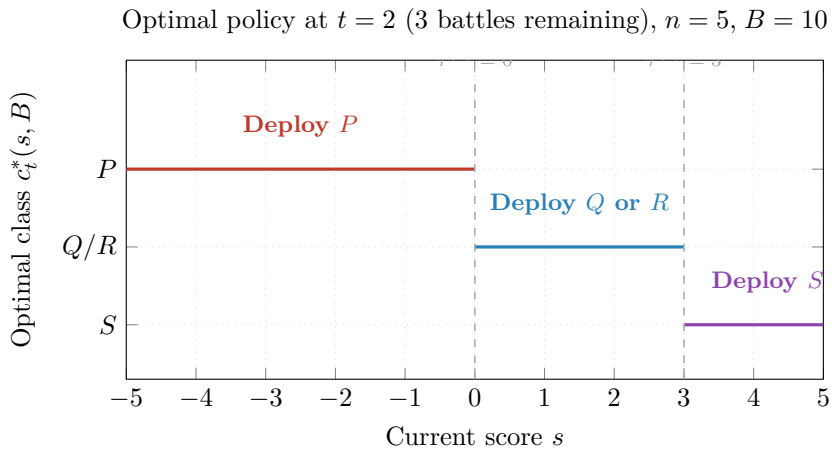

Figure 3: Optimal class deployment at $t = 2$, $n = 5$, $B = 10$, uniform weights.

Class P is deployed when trailing ($s < 0$); Q or R at intermediate scores ($0 \leq s < 3$); S when comfortably ahead ($s \geq 3$). Dashed verticals mark the thresholds $\tau^{(1)} = 0$ and $\tau^{(2)} = 3$.

11 Discussion and Conclusions

We have embedded the four-class soldier framework into a fully dynamic setting, modelling a war as a finite-horizon controlled Markov chain driven by fuzzy battle outcomes and evaluated under four families of weighted scoring rule. Three conclusions merit emphasis.

Fuzzy draws are structurally essential. The outcome $r_t = 0$ is not a degenerate edge case. A draw depletes the budget and occupies a battle slot without advancing the score. Under late-heavy weights, a drawn midgame battle can be more damaging than a loss: it consumes a slot that could have been reserved for a high-weight late encounter, where a strong class would have yielded greater expected score gain.

The threshold policy is principled, minimal, and fast. Theorem 8.2 establishes that the optimal policy has a simple two-threshold structure: deploy the cheapest class sufficient to achieve the required score gain. When trailing, invest in class P; when comfortably ahead, conserve budget with class S. The policy mirrors resource-allocation heuristics in military doctrine [4] and competitive sports strategy [3], and is computable in $O(n^2)$ time via Algorithm 1.

The CLT approximation enables rapid planning. Equation (27) gives a closed-form win-probability estimate valid for $n \geq 10$ and any fixed-class policy, supporting fast scenario analysis without invoking the full backward induction.

Future directions. Natural extensions include: (i) *adaptive enemies* that observe the deployer's class and update their configuration distribution, yielding a two-player stochastic game; (ii) *continuous attribute spaces*, in which (2) is integrated against a prior over $(H, A) \in \mathbb{R}^2_{>0}$; (iii) *infinite wars* ($n \to \infty$), where the deployer maximises the time-average score and ergodic MDP theory applies; and (iv) *coalition warfare*, in which a team of soldiers from mixed classes cooperates within a single battle, extending the product t-norm to an n-ary aggregation operator.

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Glossary

Attribute kind (F, ω) The binary classification of a soldier or enemy attribute: F (any finite positive value) or $\omega = +\infty$ (transfinite).

Backward induction The dynamic-programming algorithm that computes the optimal value function by propagating terminal rewards backwards in time, one step per iteration, via the Bellman recursion (17).

Bellman recursion The functional equation (17) characterising the optimal value function of a finite-horizon MDP; the computational basis for Algorithm 1.

Budget (B) The total resource pool available for an entire war; reduced by ρ_c whenever class c is deployed.

Centroid defuzzification The mapping of a fuzzy output distribution to a crisp scalar by computing its probability-weighted mean; see (6).

CLT approximation The normal approximation (27) to the war win probability, obtained by applying the Lindeberg CLT to the independent weighted increments $\{w_t R_t\}$.

Discrete concavity The property (19): consecutive score gains yield diminishing increases in win probability. This is the key structural property exploited in Theorem 8.2.

Draw ($r_t = 0$) A battle outcome in which neither side prevails; the score is unchanged, but one budget unit and one battle slot are consumed.

Fuzzy membership function (μ) A function mapping a crisp input to a degree of membership in $[0, 1]$; here the sigmoid σ_α smoothly interpolates between crisp win and loss.

Gambler's Ruin The classical problem of a binary random walk with absorbing barriers at $\pm k$; models the first-to-k war variant under an all-S policy via (24).

Markov Decision Process (MDP) A sequential decision problem specified by a state space, action space, transition kernel, and reward function; see Definition 6.1.

Policy (π) A sequence of decision rules $\pi_t(s, b)$ mapping current score and remaining budget to a class choice; the object optimised in (15).

Sharpness parameter (α) Controls the width $O(1/\alpha)$ of the sigmoid transition zone; as $\alpha \to \infty$ the boundary becomes crisp.

Sigmoid (σ_α) The function $(1+e^{-\alpha x})^{-1}$; the canonical smooth step function used to model gradational battle outcomes.

Soldier class (P, Q, R, S) One of the four archetypes defined by $(\kappa_h, \kappa_a) \in \mathbf{2}^2$; see (1) and the companion treatise.

t-norm A binary aggregation operator on $[0, 1]$ satisfying commutativity, associativity, monotonicity, and boundary conditions; used to combine independent fuzzy membership degrees.

Threshold policy The optimal two-threshold structure of Theorem 8.2: deploy stronger (costlier) classes when the current score is below $\tau^{(1)}$, weaker classes above $\tau^{(2)}$.

Value function (V_t^*) The maximum achievable probability of winning the war from state (t, s, b) under the optimal policy; satisfies (17).

War outcome (W) The sign of the terminal weighted score: $+1$ (victory), 0 (drawn war), or -1 (defeat); see (13).

Weighted scoring rule An assignment of positive weights w_t to each battle; four canonical profiles are listed in Table 1.

The End